بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

In the name of Allah, the Most Gracious, the Most Merciful

Introduction

Presented By

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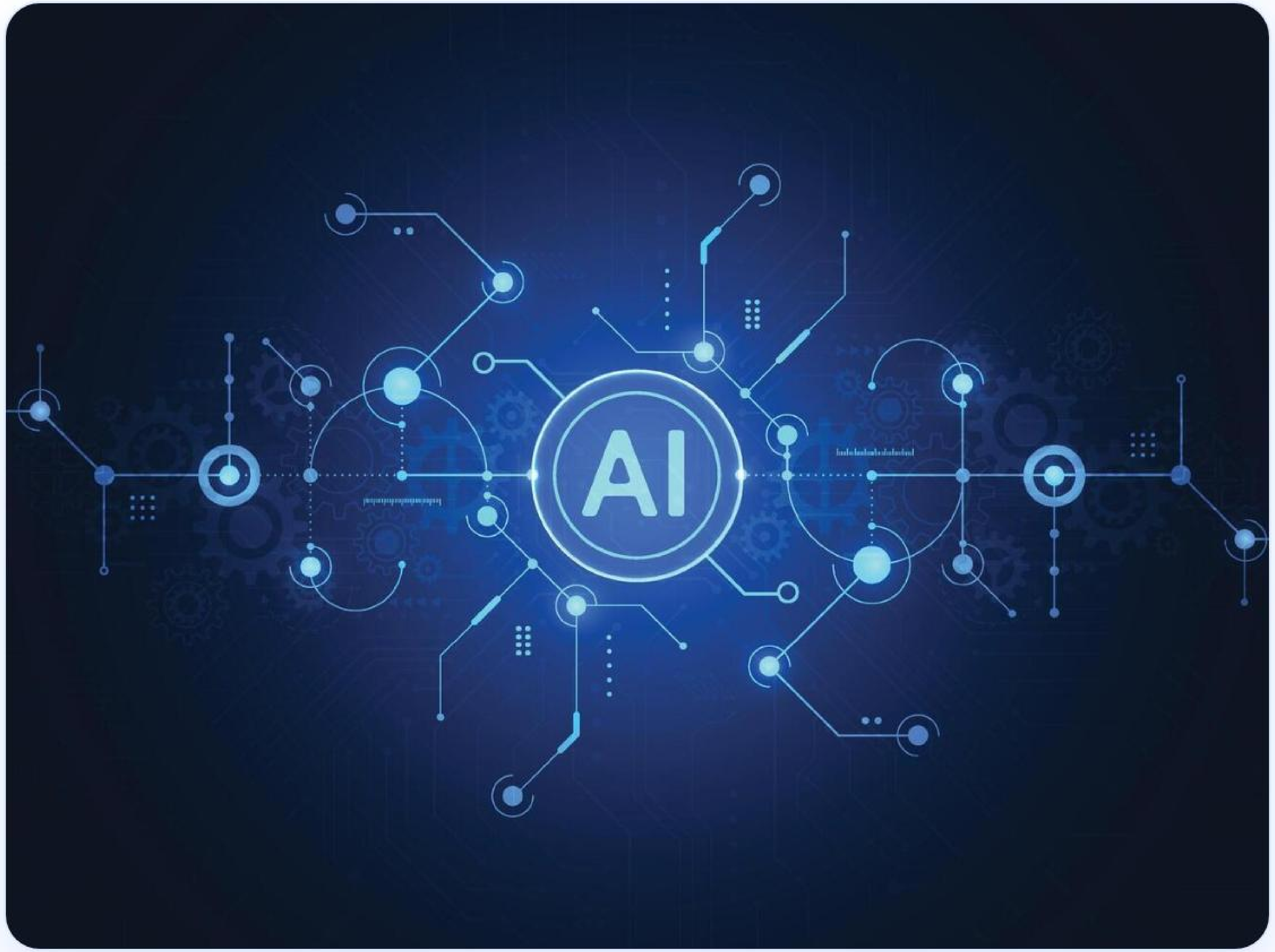
Semester:	**3rd Semester**



WEEK 12

Introduction to Machine Learning Types, Classification and Regression

What is Machine Learning?

A core branch of Artificial Intelligence (AI).

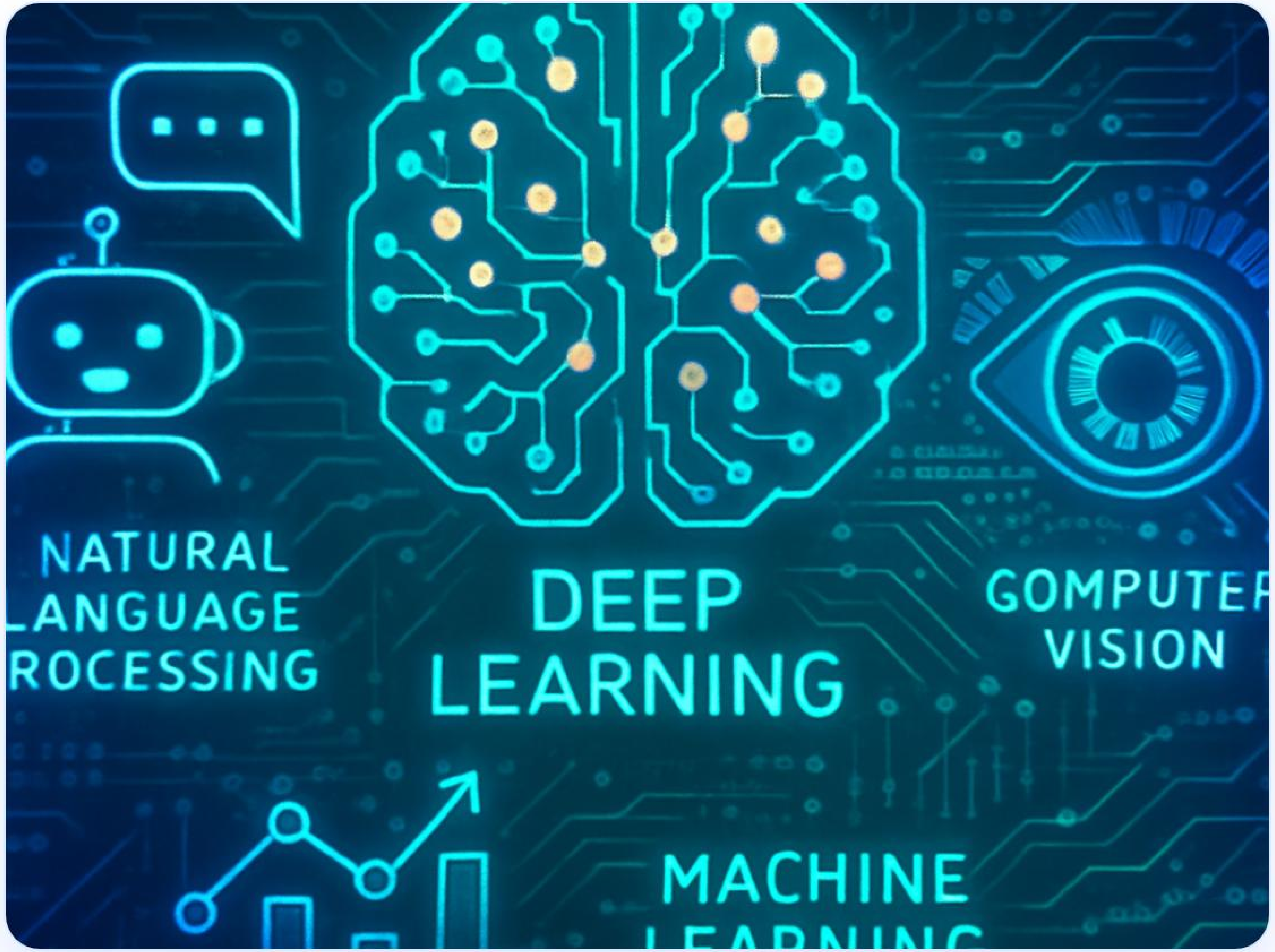
It enables computers to **learn from data** and make predictions or decisions without being explicitly programmed.

Main Types of Machine Learning:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Real-World Examples:

- Netflix Recommendations
- Face Recognition
- Spam Email Detection



Classification

Used to place data into distinct categories or discrete classes.

Key Characteristics: The output generated is always a discrete **label or category**.

Common Examples:

Email Filter

Spam or Not Spam

Image Recognition

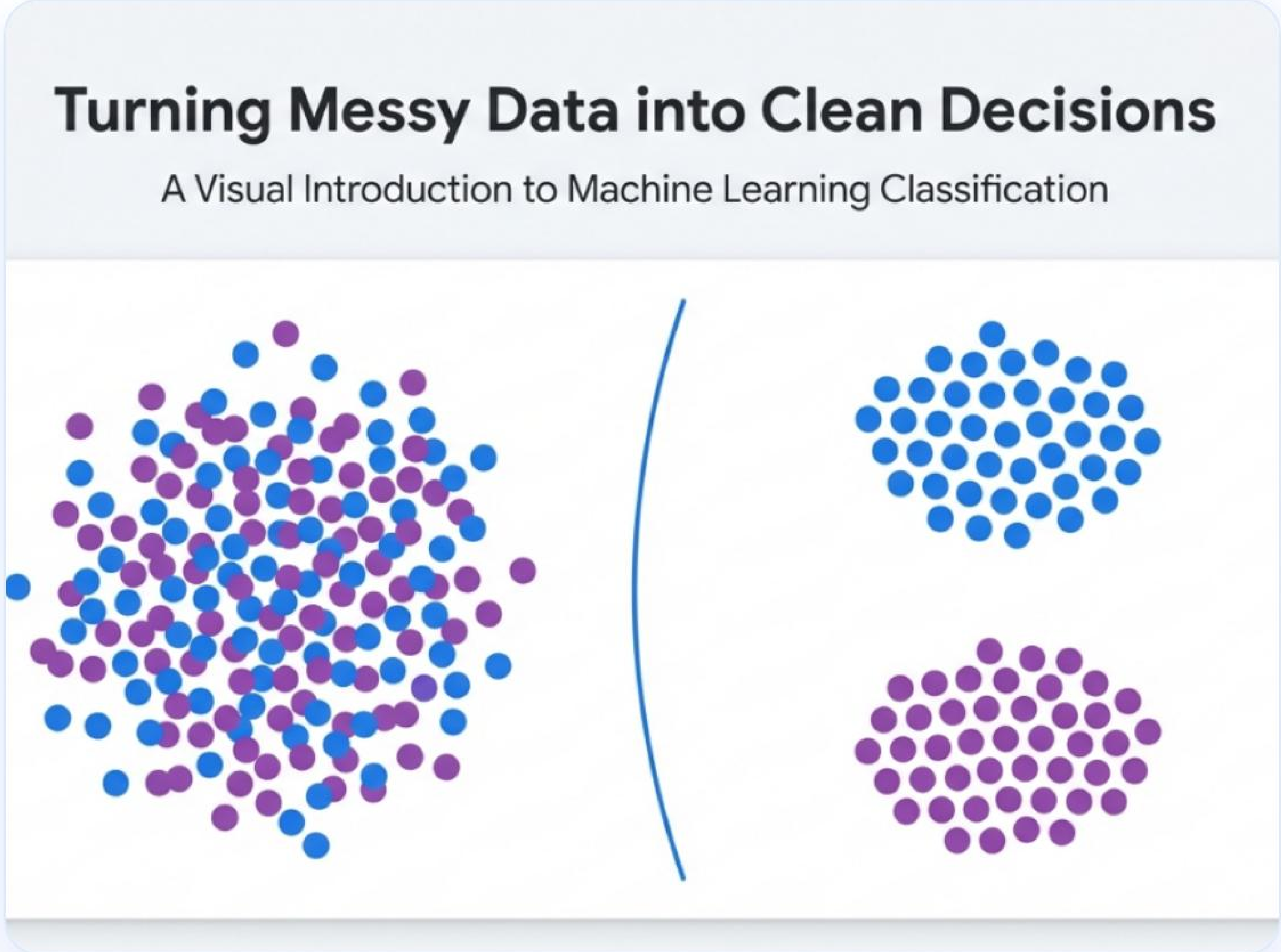
Cat or Dog Image

Academic Status

Student Pass or Fail

Medical Test

Disease Positive or Negative



Classification predicts categories, not numbers.

Regression

Used to predict continuous numerical values.

Key Characteristics: The output is always a continuous **numeric value**.

Common Examples:

Real Estate

House Price Prediction

Weather Forecasting

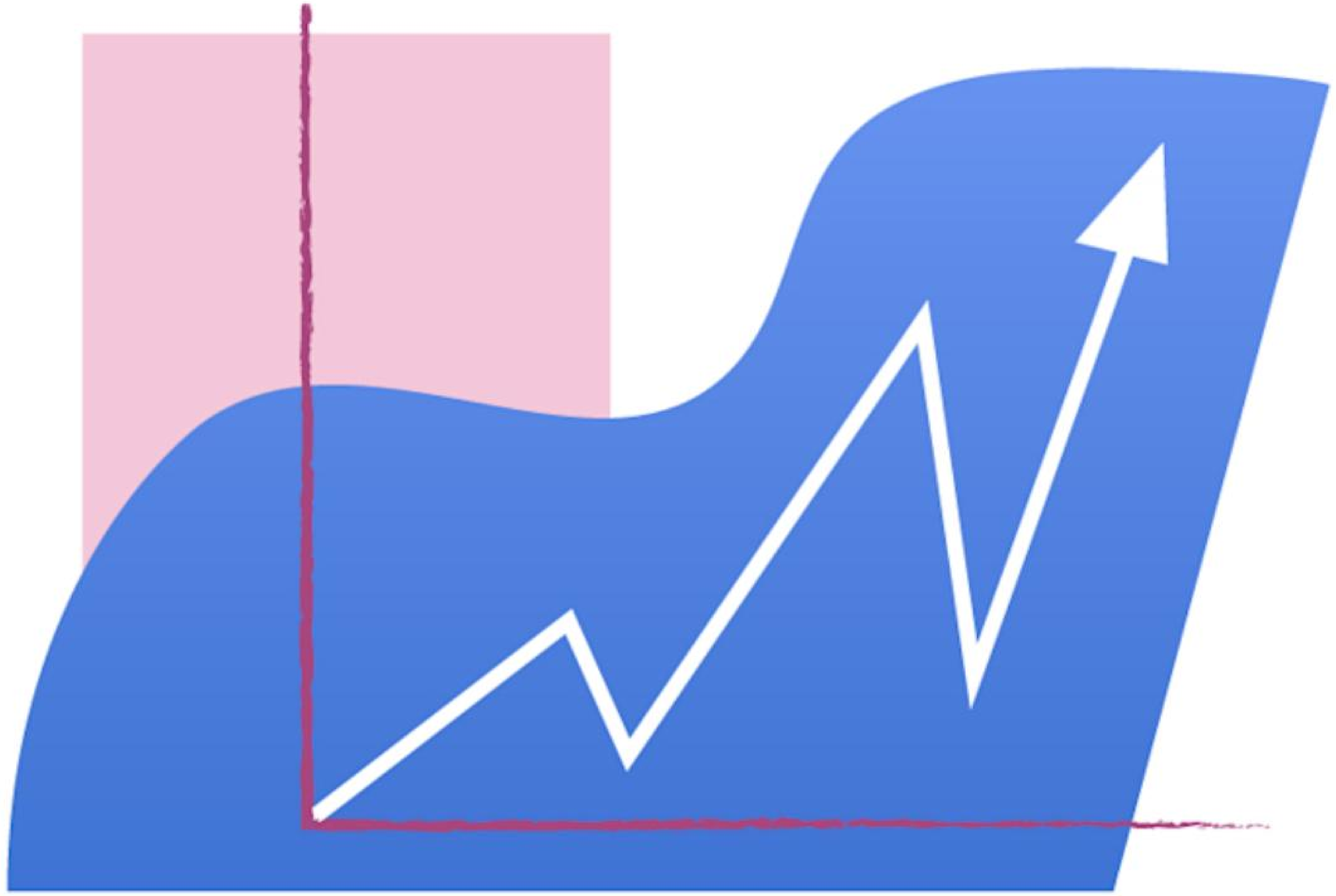
Temperature Prediction

Academics

Student Marks Prediction

Business Analytics

Sales & Revenue Prediction



Regression predicts numbers instead of categories.

Classification vs Regression

Classification	Regression
Primary Goal Predicts Categories or Discrete Classes	**Primary Goal** Predicts Continuous Numerical Values
Output Type Categorical Label (e.g., Yes/No, Red/Blue)	**Output Type** Numeric Value (e.g., $250,000, 32°C)
Typical Example Spam Email Detection	**Typical Example** House Price Prediction

Classification = Category | Regression = Number

Conclusion

- **Machine Learning** helps computers learn patterns directly from data.
- **Classification** accurately categorizes data into discrete classes or labels.
- **Regression** forecasts continuous numerical values.
- Both are essential core techniques used extensively in modern **Artificial Intelligence**.

Thank You

Image Sources

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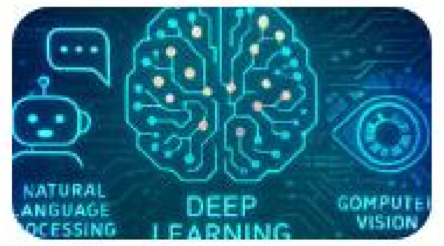

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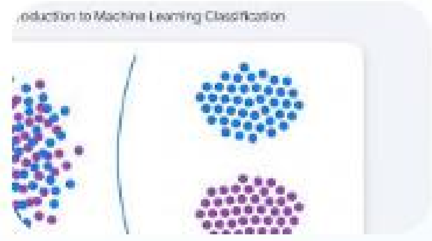

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